

PCAP (Python Institute) – Practice Exam

Duration: 2 Hours

Total Points: 100

Instructions

- Answer all questions.
 - No external libraries unless explicitly stated.
 - You may assume Python 3.11 behavior.
 - Write clear, idiomatic Python.
 - Partial credit applies for coding questions.
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Section A – Multiple Choice Questions (30 points)

Each question: 2 points

Q1

What is the output of the following code?

```
x = [1, 2, 3]
y = x
x.append(4)
print(y)
```

A. [1, 2, 3] B. [1, 2, 3, 4] C. [4] D. Error

Q2

What is the result of `3 // 2` in Python? A. 1.5 B. 1 C. 2 D. 1.0

Q3

Which of the following is immutable? A. list B. dict C. set D. tuple

Q4

What does this expression return?

```
bool("False")
```

A. False B. True C. None D. Error

Q5

What is the output?

```
print(2 ** 3 ** 2)
```

A. 64 B. 512 C. 256 D. 36

Q6

Which keyword is used to handle exceptions? A. catch B. try C. except D. handle

Q7

What is the result of:

```
len({1, 1, 2, 3})
```

A. 4 B. 3 C. 2 D. Error

Q8

What does `is` compare in Python? A. Value equality B. Memory identity C. Type equality D. String content

Q9

What is the output?

```
print("abc" * 2)
```

A. abc2 B. abcab C. abc abc D. Error

Q10

Which data structure does NOT allow duplicate keys? A. list B. tuple C. dict D. set

Q11

What is the output?

```
x = 10
print(type(x) == int)
```

A. True B. False C. int D. Error

Q12

Which method adds a single element to a list? A. add() B. append() C. extend() D. insert()

Q13

What is the result?

```
print(0.1 + 0.2 == 0.3)
```

A. True B. False C. Error D. None

Q14

Which of the following creates a generator? A. [] B. () C. {} D. yield

Q15

What is the output?

```
print(list(range(3)))
```

A. [1, 2, 3] B. [0, 1, 2] C. [0, 1, 2, 3] D. [1, 2]

Section B – Code Output Prediction (25 points)

Each question: 5 points

Q16

```
def f(x, lst=[]):  
    lst.append(x)  
    return lst  
  
print(f(1))  
print(f(2))
```

Q17

```
for i in range(3):  
    if i == 1:  
        continue  
    print(i, end=" ")
```

Q18

```
x = [i*i for i in range(4)]  
print(x[-1])
```

Q19

```
a = {1: "A", 2: "B"}  
print(a.get(3, "C"))
```

Q20

```
try:  
    print(10 / 0)  
except ZeroDivisionError:  
    print("Error")  
finally:  
    print("Done")
```

Section C – Short Answer (15 points)

Each question: 5 points

Q21

Explain the difference between a list and a tuple in Python.

Q22

What is Python's LEGB rule?

Q23

Explain the purpose of the `with` statement.

Section D – Coding Tasks (30 points)

Q24 (10 points)

Write a function `is_palindrome(s: str) -> bool` that returns True if the string is a palindrome ignoring case.

Q25 (10 points)

Write a function that takes a list of integers and returns a dictionary with:

- keys: integers
- values: frequency count

Example:

```
[1,1,2,3,3,3]
```

Output:

```
{1:2, 2:1, 3:3}
```

Q26 (10 points)

Write a class `BankAccount` with:

- attribute `balance`
 - method `deposit(amount)`
 - method `withdraw(amount)` (prevent overdraft)
 - method `get_balance()`
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End of Exam

Bonus Challenge (Optional)

Implement a generator that yields Fibonacci numbers indefinitely.